Durability (ACID Property)

1. Overview

Durability means once a transaction is **successfully committed**, its changes must be **permanent**.

```None
Commit successful

→ Data must survive crashes, restart, power failure
```

Even if the system fails immediately after commit, committed data should still exist after recovery.

2. Simple Meaning

```None
If transaction says SUCCESS,

data cannot disappear later.
```

3. Banking Example

Transfer ₹50 from Account A to Account B.

Before:

```None
A = 500

B = 300
```

After successful commit:

```None
A = 450

B = 350
```

If server crashes 1 second later, after restart balances must remain:

```None
A = 450

B = 350
```

Not old values.

4. Why Durability Matters

Without durability:

- Payments disappear
- Orders vanish after confirmation
- Bookings lost
- User data missing
- Audit records unreliable

5. Commit Means Promise

When DB returns:

```None
COMMIT SUCCESS
```

It guarantees:

- Data written safely
- Recoverable after crash
- Visible after restart

6. How Databases Achieve Durability

A. Write-Ahead Logging (WAL)

Changes written to log before final commit.

Used by:

- PostgreSQL
- MySQL

B. Disk Persistence

Store data on SSD/HDD, not only RAM.

C. Crash Recovery

On restart:

```None
Replay logs

Restore committed transactions

Rollback incomplete ones
```

D. Replication

Multiple copies on replicas.

7. SQL Example

```sql
BEGIN;


UPDATE accounts

SET balance = balance - 50

WHERE id='A';


UPDATE accounts

SET balance = balance + 50

WHERE id='B';


COMMIT;
```

After commit, changes must survive failure.

8. Durability vs Atomicity

Property	Meaning
Atomicity	All or nothing
Durability	Once committed, forever persisted

9. Durability vs Availability

Concept	Meaning
Durability	Data survives failure
Availability	System remains accessible

A system may be temporarily unavailable but still durable.

10. Real-World Example: Ecommerce

User pays and order confirmed.

After restart:

```None
Order must still exist

Payment record must still exist

Inventory deduction must persist
```

11. Cloud Examples

Durability often improved using:

- Replicated databases
- Snapshots
- Multi-zone storage
- Managed services like Amazon RDS

12. Trade-offs

Stronger durability may increase:

- Write latency
- Replication overhead
- Disk sync cost

Some systems trade durability for speed temporarily.

Example:

```None
write to memory first

flush later
```

Used in some caches.

13. Interview Answer Example

Q: What is durability?

> Durability guarantees that once a transaction commits successfully, its data remains permanently stored and recoverable even after crashes or restarts.

14. Short Revision Notes

```None
Durability =

Committed data survives failures



Techniques:

WAL

disk storage

replication

recovery logs

backups
```

15. Crash Scenario

```None
Commit success

Power failure

Restart

Data still present
```

That is durability.

16. One-Line Summary

Durability ensures that successfully committed transactions are permanently stored and remain intact even if the system crashes afterward.